

**Data-Driven Prediction of Formation Energy and Structure
in CoAu Nanoparticles Using Morphological and
Coordination Features**

A PROJECT REPORT

Submitted to

Amrita Vishwa Vidyapeetham

in partial fulfilment for the award of the degree of

By

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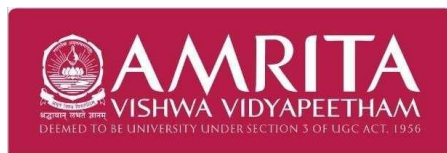
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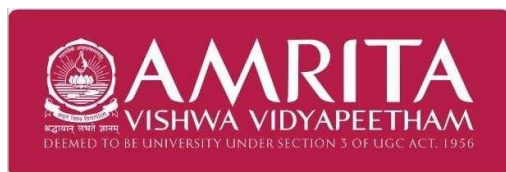


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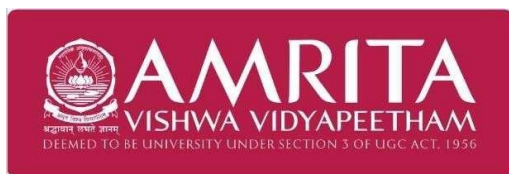
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ABSTRACT

The design and optimization of bimetallic Cobalt-Gold (CoAu) nanoparticles are pivotal for advancements in catalysis and clean energy technologies. However, traditional simulation methods like Density Functional Theory (DFT) are computationally expensive, creating a bottleneck in materials discovery. This project addresses this challenge by developing a robust data-driven framework to predict key energetic properties of CoAu nanoparticles. From a comprehensive dataset of over 156,000 configurations, we utilized a representative subset of 15,071 samples for model training and validation. A multi-output deep neural network was implemented to simultaneously predict both formation energy and total energy, achieving exceptional R-squared (R^2) scores of 0.99788 and 0.99945, respectively. A comparative Random Forest model yielded slightly higher performance, further validating the high accuracy of this machine learning approach. This work successfully demonstrates that data-driven models can rapidly and reliably predict nanoparticle energetics, significantly accelerating the virtual screening process for designing novel materials.

Keywords: Nanoparticles, Cobalt-Gold (CoAu), Formation Energy, Machine Learning, Deep Neural Network, Random Forest, Materials Informatics, Predictive Modeling.

ACKNOWLEDGEMENT

This project work would not have been possible without the contribution of many people. It gives me immense pleasure to express my profound gratitude to our honorable Chancellor **Sri Mata Amritanandamayi Devi**, for her blessings and for being a source of inspiration. I am indebted to extend my gratitude to our Director, **Mr. I B Manikantan** Amrita School of Computing and Engineering, for facilitating us all the facilities and extended support to gain valuable education and learning experience.

I register my special thanks to **Dr. V. Jayakumar**, Principal, Amrita School of Computing and Engineering for the support given to me in the successful conduct of this project. I wish to express my sincere gratitude to my supervisor **Dr. J JEYALAKSHMI**, Assistant Professor, **Dr. S. Sountharajan**, Chairperson & **Dr. S. Natarajan**, Program Chair, Department of Computer Science and Engineering, for his inspiring guidance, personal involvement and constant encouragement during the entire course of this work.

I am grateful to Project Coordinator, Review Panel Members and the entire faculty of the Department of Computer Science & Engineering, for their constructive criticisms and valuable suggestions which have been a rich source to improve the quality of this work.

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LIST OF SYMBOLS AND ABBREVIATIONS

CoAu	Cobalt-Gold
CSIRO	Commonwealth Scientific and Industrial Research Organisation
DFT	Density Functional Theory
E _{form}	Formation Energy
E _{total}	Total Energy
GCN	Generalized Coordination Number
GUI	Graphical User Interface
ANN	Artificial Neural Network
DNN	Deep Neural Network
GNN	Graph Neural Network
ML	Machine Learning
MLP	Multi-Layer Perceptron
NN	Neural Network
PCA	Principal Component Analysis
ReLU	Rectified Linear Unit
RF	Random Forest
MAE	Mean Absolute Error
MSE	Mean Squared Error
R ²	R-squared Score (Coefficient of Determination)

CHAPTER-1

INTRODUCTION

1.1 Background and Motivation

Bimetallic nanoparticles, particularly those composed of Cobalt and Gold (CoAu), are at the forefront of materials science research due to their unique catalytic, magnetic, and optical properties. These advanced materials are pivotal for developing next-generation technologies in fields such as clean energy, including fuel cells and hydrogen production, and industrial catalysis for chemical manufacturing. The precise atomic arrangement, size, and composition of these nanoparticles determine their stability and functional performance.

However, the approach to discovering and refining novel nanoparticles is a significant challenge. Traditional research methods rely on a combination of physical experiments and computationally intensive simulations, such as density functional Theory. While accurate, these methods are extremely time-consuming and resource-intensive, creating a major bottleneck that slows the pace of innovation. There is a pressing need for faster, more efficient methodologies to predict the properties of nanoparticles. This project is motivated by the potential of machine learning to bridge this gap, leveraging existing simulation data to create predictive models that can assess nanoparticle stability in a fraction of the time, thereby accelerating the design and discovery cycle for new materials.

1.2 Problem Statement

The core problem this project addresses is the inefficiency of traditional methods for determining the stability of bimetallic CoAu nanoparticles. Key properties like formation energy are critical indicators of a nanoparticle's stability and viability for practical applications, but their calculation is computationally demanding. This computational expense makes the large-scale screening of thousands of potential

nanoparticle candidates impractical, hindering the rational design of materials with tailored properties.

The current workflow for material discovery is therefore reactive and slow. A significant gap exists between the vast number of potential nanoparticle configurations and our capacity to evaluate them efficiently. This problem aims to find accurate estimating the formation energy and total energy of CoAu nanoparticles based on their structural and physicochemical descriptors alone, without the need for repeated, time-consuming simulations.

1.3 Project Objectives

To address the problem statement, the following specific objectives were established for this project:

- To preprocess and analyze the CoAu nanoparticle dataset, preparing a large set of structural and energetic features for model training.
- To develop, train, and validate a multi-output deep neural network (NN) capable of simultaneously predicting the Formation Energy and Total Energy of nanoparticles.
- To implement a Random Forest (RF) regression model to serve as a robust baseline for evaluating the performance of the neural network.
- To do a comparative analysis of the model results to determine the effectiveness of a deep learning approach versus a traditional ensemble method for this specific materials science problem.

1.4 Scope and Limitations

The scope of this project is precisely Clearly delineated to examine applications of

ml for predicting the properties of a specific nanomaterial system. The research is exclusively focused on bimetallic Cobalt-Gold (CoAu) nanoparticles, utilizing a pre-existing dataset generated from computational simulations. All models are trained and validated using the features provided within this dataset, which include a wide range of structural and energetic descriptors. The predictive targets of the models are strictly limited to the two energetic properties available in the dataset: Formation Energy and Total Energy. Any other properties, such as magnetic moments or catalytic activity, are considered outside the scope of this investigation.

The methodological approach is centered entirely on supervised machine learning, with a specific focus on regression tasks. Two distinct modeling techniques are implemented and compared: a deep neural network, selected for its effectiveness in handling complex non-linear interactions and a RF regressor, selected as a powerful and widely-used ensemble baseline. The project encompasses all stages of the machine learning workflow, from data preprocessing and feature scaling to model training, validation, and performance evaluation using standard metrics like MAE and R^2 score. The final output is a comparative analysis of these two models based on their predictive accuracy for the specified targets.

A primary limitation of this project is its complete reliance on pre-existing simulation data. The models are trained to recognize patterns within this data, and their predictive accuracy is therefore inherently bound by the quality, accuracy, and comprehensiveness of the original simulations. The project does not involve the generation of new first-principles data through methods like Density Functional Theory (DFT), nor does it include the physical synthesis or experimental validation of CoAu nanoparticles. Consequently, the findings represent a validation of a predictive methodology rather than a fundamental discovery of new physical phenomena.

Furthermore, the models developed are fundamentally "black box" in nature; they are designed for predictive tasks and do not provide deep insights into the

underlying physics governing nanoparticle stability. While they can accurately forecast energetic properties based on structural inputs, they do not explain the causal relationships in a way that a first-principles simulation does. The project also concludes upon the evaluation of the models and does not extend to their deployment into a live, production-level application or a user-friendly tool. As such, the practical application of these models by other researchers would require additional development works

CHAPTER-2

LITERATURE REVIEW

.2.1 Nanoparticles in Materials Science

Nanoparticles, defined as particles with at least one dimension in the 1 to 100 nanometer range, represent a revolutionary class of materials. Their significance stems from a fundamental principle: as a particle's size decreases, its surface-area-to-volume ratio increases dramatically. For instance, in a bulk material, the fraction of atoms on the surface is negligible. However, in a nanoparticle of just a few nanometers, a significant percentage of the atoms reside on the surface. This high proportion of surface atoms, along with unique quantum phenomena, gives rise to catalytic, electronic, and optical properties that are fundamentally different from those of their bulk counterparts.

The unique behavior of nanoparticles is governed by two primary factors. The first is quantum confinement, which occurs when the particle's dimensions are small enough to restrict the movement of electrons, causing their energy levels to become discrete rather than continuous. This directly alters the electronic band structure and leads to size-dependent properties, such as the fluorescence of quantum dots or the surface plasmon resonance in noble metal nanoparticles like gold, which is responsible for their vibrant colors. The second factor is the prevalence of highly active surface sites. The atoms at the surfaces, edges, and corners of a nanoparticle have lower coordination numbers and are more chemically reactive than atoms within a bulk crystal, making them exceptional sites for catalysis.

Bimetallic nanoparticles, which integrate two distinct metallic elements into a single particle, offer an even more sophisticated platform for materials design. They allow for the creation of materials with functionalities that are not accessible with single-element nanoparticles, often due to synergistic effects. This synergy arises from the

interaction between the two elements, where one metal can favorably modify the electronic structure or geometric arrangement of the other, leading to enhanced catalytic activity or stability. The characteristics of bimetallic nanoparticles can be finely tuned by controlling their size, overall composition, and internal atomic arrangement, which may result in structures such as core-shell, Janus (two-faced), or intermixed alloys.

The Cobalt-Gold (CoAu) bimetallic system, which is the focus of this study, is a prime example of a multifunctional nanomaterial. Cobalt is known for its excellent magnetic properties and catalytic activity in important industrial reactions, while gold is valued for its extreme chemical stability and unique plasmonic properties. By combining them, it is possible to design, for example, a magnetically separable catalyst that is both highly active and resistant to degradation. However, the performance of a CoAu nanoparticle is critically dependent on its precise structure. The vast design space—encompassing a nearly infinite number of possible sizes, compositions, and atomic arrangements (isomers)—makes the identification of the most stable and functional structures a monumental scientific challenge, one that cannot be tackled by experimental trial-and-error alone.

2.2 Computational Methods in Materials Discovery (DFT)

The Computational modeling has become an indispensable third pillar of modern science, standing alongside theory and experimentation. Within materials discovery, computational methods allow researchers to predict and analyze the behavior of materials at the atomic level, providing insights that can guide experimental efforts. The most powerful of these are first-principles (or *ab initio*) methods, which derive their results directly from the fundamental laws of quantum mechanics without relying on empirical data. Among these, Density Functional Theory-DFT has become the most widely adopted method, offering a favorable balance between accuracy and computational efficiency

The central concept of DFT is a revolutionary simplification of the quantum

mechanical many-body problem. Rather than trying to compute the complex wavefunction of a system, which depends on the coordinates of every electron, DFT is based on the concept that all ground-state properties can be uniquely determined from the system's electron density. Since electron density depends only on three spatial coordinates, this greatly simplifies the problem. In practical applications, the Kohn-Sham approach to DFT is employed, transforming the problem into a set of single-electron equations. These equations provide an accurate description of the energy and electronic structure of atoms, molecules, and solids, enabling precise calculation of the system's total energy (E_{total})

From the DFT-calculated total energy, researchers can derive critical properties that describe a material's stability and viability. A key metric for alloys and compounds is the formation energy (E_{form}), which quantifies the energy change when a structure is formed from its constituent elements. For a bimetallic CoAu nanoparticle, it is calculated as:

$$E_{\text{form}} = E_{\text{total}}(\text{Co}_x\text{Au}_y) - [x \cdot E_{\text{bulk}}(\text{Co}) + y \cdot E_{\text{bulk}}(\text{Au})]$$

Here, $E_{\text{total}}(\text{Co}_x\text{Au}_y)$ is the total ground-state energy of the nanoparticle as calculated by DFT, while $E_{\text{bulk}}(\text{Co})$ and $E_{\text{bulk}}(\text{Au})$ are the reference energies of single atoms within their most stable bulk crystalline structures. A negative formation energy indicates that the nanoparticle is thermodynamically stable and its formation is an exothermic process (it releases energy), making it likely to exist. Conversely, a positive formation energy suggests the structure is unstable and would tend to decompose back into its separate elemental forms.

Despite its predictive power, DFT suffers from a major drawback: its high computational cost. The time required for a DFT calculation scales non-linearly, often cubically ($O(N^3)$) or worse, with the number of atoms (N) in the system. This scaling arises from the need to solve the Kohn-Sham equations, which involves mathematical operations on large matrices whose size is proportional to N . This means that doubling

the number of atoms in a nanoparticle could increase the computation time by a factor of eight or more. This computational bottleneck makes DFT impractical for the high-throughput screening of thousands of potential nanoparticle candidates or for modeling very large systems, severely limiting the speed at which new materials can be discovered.

2.3 Machine Learning for Predicting Material Properties

To overcome the computational bottleneck of first-principles methods like DFT, Materials science has increasingly adopted a data-driven approach, commonly described as the ‘fourth paradigm’ of scientific discovery, alongside empirical, theoretical, and computational methods. The central principle of this paradigm is to leverage machine learning (ML) to learn the complex and often non-linear relationships that connect a material's structure to its properties. Instead of running a new, time-consuming simulation for every candidate material, a model is trained on a large, existing dataset of pre-computed results. Once trained, this model acts as a surrogate, capable of making predictions for new, unseen structures almost instantaneously, thereby drastically reducing the reliance on direct simulation and enabling high-throughput virtual screening.

The machine learning workflow in materials science follows a structured process. It begins with data acquisition, where a comprehensive dataset is assembled. For this project, this consists of thousands of CoAu nanoparticle structures and their corresponding formation energies calculated via DFT. The next critical step is feature engineering, where each atomic structure is converted into a fixed-length numerical vector, or "descriptor." This vector must effectively capture the key structural and chemical information that governs the material's properties. The core of the process is model training, where the ML algorithm iteratively learns from the feature vectors and their associated true energy values. During training, the model adjusts its internal parameters to minimize a pre-defined loss function, which quantifies the error between its predictions and the actual DFT-calculated values. The model's ability to generalize is then assessed in the validation phase, where its performance is measured

on a separate, unseen portion of the dataset (the test set) to ensure it can make accurate predictions on new data.

The task of predicting a continuous value, such as formation energy, is known as a regression problem. This is distinct from a classification problem, where the goal would be to predict a discrete category (e.g., whether a material is metallic or insulating). In regression, the model's performance is measured by how close its predictions are to the true continuous values. The primary tool for guiding the training process is the loss function. A common and effective choice for regression is the Mean squared Error, given by the formula

$$\text{MSE} = (1/n) * \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

In this equation, n represents the total number of samples in the dataset, Y_i is the true, DFT-calculated energy for a given nanoparticle, and $\hat{Y}_i$ is the energy predicted by the ML model for that same nanoparticle. The term $(Y_i - \hat{Y}_i)$ is the prediction error, or "residual." By squaring this term, two objectives are achieved: it ensures that all errors are positive so they do not cancel each other out. The training process, therefore, systematically adjusts the model's parameters to find the configuration that minimizes this average squared error across the entire dataset. By doing so, the model effectively learns a highly accurate mapping from a nanoparticle's structural features to its energetic properties.

2.4 Existing Models for Nanoparticle Energy Prediction

The success of any machine learning model in materials science is contingent on two synergistic components: first, the transformation of a physical atomic structure into a meaningful numerical representation (feature engineering), and second, the selection of a suitable learning algorithm (model architecture) capable of mapping those features to the target property.

Feature Representation: Translating Structure into Numbers

A primary challenge in materials informatics is that machine learning algorithms operate on numerical vectors, not on 3D atomic coordinates. The process of feature engineering, therefore, involves creating a numerical "fingerprint" or "descriptor" for each nanoparticle that encodes its essential structural and chemical information. The literature reveals a hierarchy of feature types, ranging from simple to highly complex:

Compositional and Structural Features: The simplest descriptors are derived from the material's overall composition and geometric structure. These include stoichiometric ratios (e.g., the percentage of Co vs. Au atoms), elemental properties (e.g., electronegativity, atomic mass), total atom count, average radius, volume, and surface area. While easy to compute, these features often lack the detailed information about the local atomic environment that governs energetic stability.

Coordination-Based Features: To capture the local environment, descriptors based on atomic coordination are crucial. These features quantify the number of nearest neighbors for each atom, providing insight into bonding and surface effects. A particularly powerful descriptor highlighted in the literature is the Generalized Coordination Number (GCN).

$$GCN_i = \sum_{j \in \text{neighbors of } i} CN_j$$

This provides a richer, second-order description of the local topology, which has been shown to be highly correlated with surface reactivity and energy.

Advanced Representations for Automated Learning: State-of-the-art methods increasingly focus on representations that allow the model to learn features automatically. "Graph Neural Networks (GNNs), for example, model a nanoparticle

as a graph, with atoms as nodes and bonds as edges. This enables the model to automatically learn effective representations from atomic connectivity, eliminating the need for manual feature engineering. Alternatively, structures can be represented as 3D voxel grids for analysis using Convolutional Neural Networks (CNNs)

Model Architectures: Choosing the Right Learning Algorithm

Once the features are defined, an appropriate algorithm must be selected. The "No Free Lunch" theorem in machine learning suggests that no single model is universally best for all problems. The choice depends on dataset size, dimensionality, and the complexity of the underlying physics. The literature on materials prediction showcases three main families of models:

Kernel and Linear Models: For smaller datasets or problems with clearer linear correlations, models like (SVR) and (KRR) are effective. They are computationally efficient and serve as excellent baselines for performance comparison.

Ensemble Methods: This family of models, which includes Random Forest and Gradient Boosted Trees, is exceptionally powerful for tabular data, like the feature matrix of a nanoparticle dataset. These techniques combine the outputs of multiple simple 'decision trees' to generate a single, highly accurate prediction. Random Forest constructs trees independently and averages their results, whereas Gradient Boosting builds trees sequentially, with each new tree aiming to correct the errors of its predecessors. Their robustness and strong performance make them widely used in materials informatics.

Deep neural Network: For large, high-dimensional datasets with intricate, non-linear relationships, DNNs are often the preferred choice. The most common architecture is the Multi-Layer Perceptron (MLP), a fully-connected network that can approximate any continuous function, making it a universal tool for regression. More advanced architectures like the aforementioned GNNs are specifically designed for structured data like molecules and represent the cutting edge of the field, capable of achieving

state-of-the-art results by learning directly from the material's graph structure.

2.5 Research Gaps and Project Justification

A thorough review of the existing literature reveals several critical gaps. These gaps highlight an opportunity for a focused investigation that can contribute valuable insights to the field. This project is specifically designed to address these shortcomings.

Limited Scope of Investigated Material Systems: A substantial body of existing research has concentrated on computationally simpler, single-element systems (e.g., gold-only or platinum-only nanoparticles) or on basic binary alloys. While foundational, these studies do not capture the complex interplay and synergistic effects present in more intricate bimetallic nanoparticles like Cobalt-Gold (CoAu). The properties of bimetallic systems are not merely a linear combination of their components; they are governed by unique ordering phenomena, charge transfers, and strain effects. Therefore, models trained on simpler systems cannot be reliably extrapolated to predict the behavior of these more complex and technologically relevant alloys.

Discrepancy Between Bulk and Nanoscale Models: Many pioneering machine learning models in materials science were developed to predict the properties of bulk crystalline materials, which are characterized by an infinitely repeating lattice structure. These models often fail when applied to nanoparticles because they do not account for the unique physics at the nanoscale. In nanoparticles, a large fraction of atoms reside on the surface, leading to dominant surface energy contributions and finite-size effects that are absent in bulk systems. A model that does not explicitly use features describing the nanoparticle's specific morphology, surface atom arrangements, and coordination environments is unlikely to capture the properties that are critical to its stability and function.

Insufficient Comparative Analysis of Advanced Models: The literature is populated with studies that successfully apply a single, powerful machine learning algorithm, such as a deep neural network or a gradient boosting model, to a specific problem. However, there is a notable scarcity of research that performs a direct, head-to-head comparison of different state-of-the-art model families (e.g., deep learning vs. advanced ensemble methods) on the same complex, high-dimensional nanoparticle dataset. The optimal choice of model architecture is highly dependent on the dataset's characteristics. Without such direct comparisons, it remains challenging for researchers to select the most appropriate and effective modeling approach for their specific nanomaterial prediction task.

This project is therefore justified by its strategic design to directly address these identified research gaps. By focusing explicitly on the complex bimetallic CoAu system, this work moves beyond simpler, single-element studies. By utilizing a rich, nanoparticle-specific feature set that includes detailed structural and coordination-based descriptors, the models are tailored to capture the essential physics of the nanoscale. Most importantly, this project provides a direct comparative analysis between a deep neural network and a Random Forest regressor, two powerful but fundamentally different machine learning paradigms. In doing so, this work not only aims to deliver a practical framework for accelerating the design of CoAu nanoparticles but also contributes valuable insights into the relative strengths of these advanced modeling techniques for complex problems in materials informatics.

CHAPTER 3

METHODOLOGY

The methodology covers the project workflow, dataset description, data preprocessing techniques, model architectures, training procedures, and the metrics used for performance evaluation.

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3.1 Project Workflow

The project was executed following a structured workflow to ensure reproducibility and clarity, as illustrated below:

- I. Data Acquisition: The primary dataset, containing DFT-calculated properties of CoAu nanoparticles, was obtained from the CSIRO Data Access Portal.
- II. Data Preprocessing: The raw data was cleaned and prepared for modeling. This involved handling non-numeric values, scaling features to a uniform range, and splitting the data into training and testing sets.
- III. Model Development: Two distinct regression models were implemented: a baseline Random Forest Regressor and a deep neural network, each with a multi-output configuration to predict both target variables simultaneously.
- IV. Model Training: The Random Forest model was fitted to the training data, while the neural network was trained iteratively using an optimization algorithm and advanced callbacks to manage the learning process.
- V. Performance Evaluation: Both trained models were used to make predictions on the unseen test set. Their performance was rigorously assessed using MAE and R^2 score.

The methodology of this project follows a structured workflow, moving from data acquisition to the final comparative analysis of the models. Figure 3.1 provides a visual overview of this end-to-end process, highlighting key stages such as data preprocessing, the parallel development and training of both Random Forest and deep neural network models, performance evaluation on the unseen test s.et, and the final comparison of their predictive accuracy

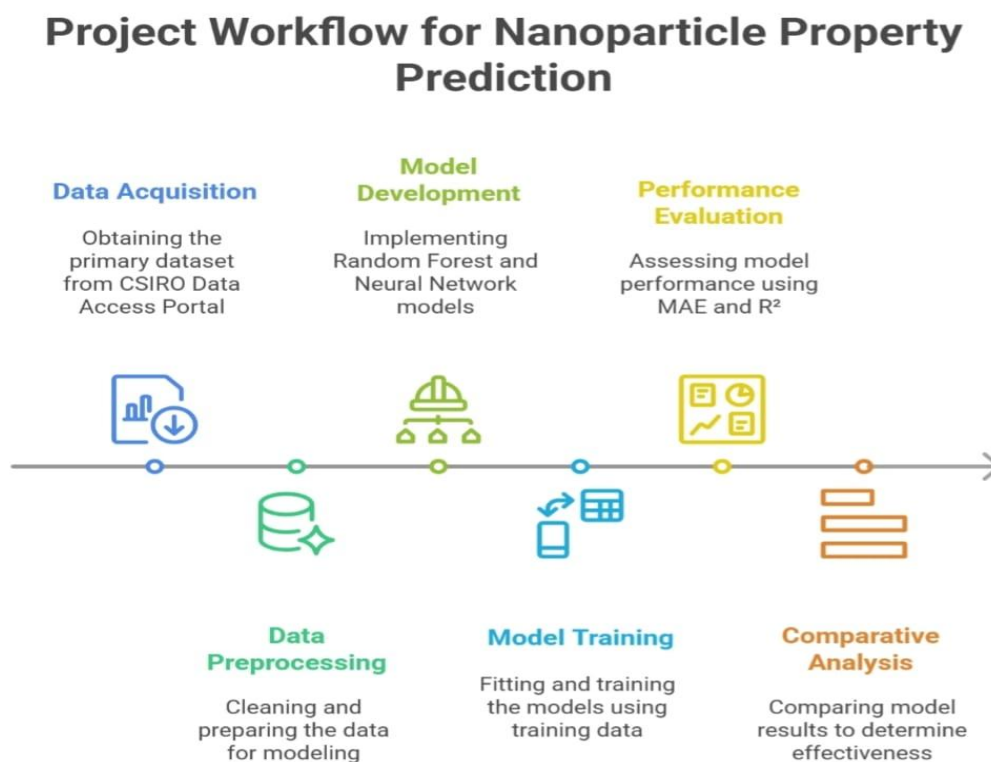


Fig 3.1 Flowchart of the End-to-End Project Methodology

3.2 Dataset Description:

The foundation of this data-driven project is the "CoAu Nanoparticle Data Set," a publicly available resource from the CSIRO Data Access Portal. This dataset was created to facilitate machine learning research in materials science by providing a

large, consistent, and high-quality set of pre-computed data from extensive computational simulations. It contains a comprehensive list of atomic and structural properties for Cobalt-Gold (CoAu) nanoparticles of varying sizes and compositions. While the full dataset contains over 156,000 configurations, this study utilized a representative subset of 15,071 nanoparticle configurations for model training, validation, and testing, a common practice for managing computational resources during iterative model development.

The dataset is characterized by its high dimensionality, featuring 1,084 input features (or descriptors) for each nanoparticle configuration. These features are designed to provide a complete numerical representation of the nanoparticle's structure and chemical environment. They can be broadly categorized as follows:

- I. **Structural Features:** These descriptors capture the overall geometry and morphology of the nanoparticle. They include basic metrics such as the total number of atoms ($N_{\text{atom_total}}$), the number of surface atoms ($N_{\text{atom_surface}}$), nanoparticle volume, and radius statistics (e.g., R_{avg} , R_{std} , R_{skew}). Additionally, this category encompasses features describing surface crystallography, including the number of atoms on $\{100\}$ and $\{111\}$ facets
- II. **Coordination Features:** This is a rich set of features describing the local atomic environment and bonding topology. It includes various types of coordination numbers (CN), such as the average CN for all atoms, for bulk atoms, and for surface atoms. Crucially, it also includes more advanced descriptors like the Generalized Coordination Number (GCN), which provides a more nuanced picture of the local chemical environment by considering the coordination of an atom's nearest neighbors.
- III. **Bonding Statistics:** This category contains detailed statistics on the chemical bonds within the nanoparticle. It includes metrics for average, standard deviation,

maximum, and minimum bond lengths and bond angles. Importantly for a bimetallic system, these statistics are provided for all possible bonding combinations: Co-Co, Au-Au, and the critical Co-Au heteroatomic bonds.

- IV. Order Parameters: These features are used to quantify the degree of local crystalline order within the nanoparticle. They include metrics such as the q6q6 spherical harmonic order parameters, which help to distinguish between different local atomic arrangements like face-centered cubic (fcc), hexagonal close-packed (hcp), or icosahedral structures.

The models developed in this project were trained to predict two key energetic properties that serve as the Target Variables. These are continuous values that are critical indicators of a nanoparticle's viability:

- I. Formation Energy (Formation_E): This is the primary target for stability prediction. It measures the thermodynamic favorability of a nanoparticle's formation relative to its constituent elements in their bulk form. A lower (more negative) formation energy indicates a more stable nanoparticle.
- II. Total Energy (Total_E): This represents the nanoparticle's total ground-state energy, as calculated by the original simulations. It is another important energetic property from which other thermodynamic quantities can be derived. pipeline.

3.3 Data Preprocessing And Feature Engineering:

Raw datasets from simulations or experiments are rarely suitable for direct input into machine learning models. They often contain non-numeric data, missing values, and features with vastly different scales. Therefore, a rigorous data preprocessing and feature engineering pipeline is a critical preliminary step to ensure the integrity of

the data and the optimal performance of the models. The following steps were systematically applied to the CoAu nanoparticle dataset.

3.3.1 Data Cleaning

The initial phase of preprocessing focused on cleaning the dataset to make it uniformly numerical and complete. The first action was to remove any columns that do not contain predictive information. In this dataset, the 'ID' column, which serves as a unique identifier for each nanoparticle configuration, was dropped as it is metadata and offers no value for predicting energetic properties.

The next step addressed the issue of missing values. Many machine learning algorithms, particularly neural networks, cannot process datasets with missing entries (often represented as NaN). To handle this, a strategy of mean imputation was employed. For each column containing missing values, the statistical mean of that column was calculated, and every missing entry within it was replaced with this calculated mean. This approach ensures that the dataset is complete and numerically consistent, allowing the models to be trained without interruption, although it is noted that this can slightly reduce the overall variance of the feature.

3.3.2 Feature Scaling

The dataset contains over 1,000 features representing a wide variety of physical properties, such as atom counts, distances (Angstroms), and energies (eV). These features exist on vastly different scales, which can hinder the performance of certain algorithms. To address this, StandardScaler was applied to all input features. This technique standardizes the data by transforming each feature to have a mean of 0 and a standard deviation of 1, according to the formula:

$$z = (x - \mu) / \sigma$$

This step is crucial for gradient-based algorithms like neural networks, as it helps the

optimization process (e.g., Adam) converge faster and more reliably. While tree-based models like Random Forest are not sensitive to feature scaling, applying a consistent preprocessing pipeline ensures that the data is prepared uniformly, allowing for a fair and direct comparison between the two model architectures. To prevent data leakage, the scaler was fitted only on the training data and was then used to transform both the training and the test sets.

3.3.3 Train-Test Split

To ensure an unbiased evaluation of the models' generalization performance, the dataset was divided into training and testing sets using an 80/20 split. Eighty percent of the data was used for model training, while the remaining 20% was reserved as an entirely unseen test set. This separation is essential for detecting potential overfitting, where a model memorizes the training data—including noise—and performs poorly on new, unseen data.

The models learn patterns exclusively from the training data. The split was performed randomly to ensure that the statistical distribution of the training set is representative of the test set and the overall dataset. For the purpose of reproducibility, a fixed `random_state` was used during this process, ensuring that the exact same split can be generated in future runs.

3.4 Model Architectures

To address the diverse set of prediction tasks—encompassing both continuous numerical values (regression) and discrete categorical labels (classification)—a dual-approach methodology was adopted. This involved the implementation of two powerful and versatile model architectures: a traditional ensemble method, Random Forest, and a deep learning method, a Deep Neural Network (DNN). Both architectures were specifically adapted to handle the different types of prediction tasks outlined for the project.

3.4.1 Baseline and Ensemble Approach: Random Forest

The Random Forest algorithm was selected for its high performance on tabular data and its ability to serve as a robust baseline. A key advantage of this ensemble method is its flexibility, allowing for its use in both regression and classification contexts with minimal changes.

Random Forest Regressor: For the regression tasks—predicting Formation Energy, Adsorption Energy, Band Gap, and Magnetic Moment—a Random Forest Regressor was used. The model builds numerous decision trees on different subsets of the dataset, and the final prediction is obtained by averaging the outputs of all trees. This averaging reduces variance and enhances the model's robustness against overfitting.

Random Forest Classifier: For the classification tasks—predicting Thermodynamic Stability (e.g., "stable" vs. "unstable") and Surface Activity (e.g., "active" vs. "inactive")—a Random Forest Classifier was used. The underlying principle is the same, but instead of averaging the outputs, the final prediction is determined by a majority vote. Each tree in the forest "votes" for a class, and the class with the most votes is assigned as the final output.

3.4.2 Deep Learning Approach: A Flexible DNN

A deep neural network was developed as a robust model to capture complex, non-linear relationships in the high-dimensional feature space. A flexible architecture was developed, consisting of a common "backbone" for feature learning and task-specific "output heads" for either regression or classification. The model was implemented using the TensorFlow and Keras libraries.

Shared Backbone Architecture: The core of the NN is a shared backbone responsible for learning a rich, hierarchical representation of the input features. This backbone consists of:

An Input Layer designed to accept the vector of 1,084 features for each nanoparticle.

Three sequential hidden blocks, each containing a Dense Layer with ReLU activation (256, 128, and 64 neurons respectively), a Batch Normalization Layer to improve training stability and speed, and a Dropout Layer (20% rate) to provide regularization and prevent overfitting.

Task-Specific Output Heads: Depending on the prediction task, a different output layer is attached to this shared backbone:

Regression Head: For predicting continuous values, a Dense output layer with a linear activation function is used. This layer contains N neurons, where N is the number of regression targets (e.g., 2 neurons for the simultaneous prediction of Formation_E and Total_E). This head is trained using a loss function suitable for regression, such as Mean Squared Error (MSE). The specific regression model trained for this report contains a total of 319,938 trainable parameters.

Classification Head: For categorical label prediction, the output layer employs a Sigmoid activation for binary classification and a Softmax activation for multi-class classification. These functions output probabilities for each class. This head would be trained using a corresponding classification loss function, such as Binary Cross-Entropy.

3.5 Training and Optimization

Training the deep neural network is an iterative optimization process in which the model's internal parameters (weights and biases) are systematically updated to capture the relationship between input features and target energy values. This process is guided by a selected optimizer and loss function, and its efficiency is further

improved through the use of intelligent callbacks.

Optimizer: Adam

For this study, the Adam (Adaptive Moment Estimation) optimizer was employed to drive the training process. Adam is a state-of-the-art, computationally efficient algorithm particularly suited for deep learning models with numerous parameters. Unlike simpler optimizers that rely on a fixed learning rate, Adam adapts the learning rate for each parameter individually by maintaining two moving averages of the gradients.”**

Other alternatives:

“The deep neural network is trained iteratively, with weights and biases updated to learn the mapping between input features and target energy values. The process relies on a chosen optimizer and loss function, with performance enhanced by intelligent callbacks. Adam, a computationally efficient optimizer, was selected for this project. Adam computes adaptive learning rates for each parameter using two moving averages of the gradients, making it ideal for models with large parameter spaces.”

“Training the network involves iterative adjustment of weights and biases to model the relationship between inputs and target energies. The Adam optimizer, chosen for its computational efficiency, uses adaptive learning rates based on gradient moving averages, improving convergence for high-dimensional models.

The first moment (the mean): This is analogous to momentum and helps to accelerate the gradient descent in the correct direction.

The second moment (the uncentered variance): This adapts the learning rate for each parameter, providing smaller updates for frequent parameters and larger updates for infrequent ones.

The choice of Adam was motivated by its proven robustness and fast convergence rates across a wide range of deep learning tasks, making it an ideal choice for training the complex neural network in this project.

Loss Function: Mean Squared Error (MSE)

The goal of training is to minimize the difference between the model's predictions and the true values in the dataset, as measured by a loss function. For this regression task, the Mean Squared Error (MSE) was used. MSE calculates the average of the squared differences between the predicted values ($\hat{Y}_i$) and the actual values (Y_i)

$$MSE = (1 / n) \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

By squaring the error term, the MSE loss function penalizes larger errors more significantly than smaller ones. This property encourages the model to avoid making substantial mistakes in its predictions. The entire training process is therefore an optimization problem where the Adam optimizer attempts to find the set of model weights that results in the lowest possible MSE value on the training data.

Callbacks for Intelligent Training

To make the training process more efficient and to prevent overfitting, two Keras callbacks were employed. These are automated functions that monitor the training process and execute pre-defined actions at the end of each epoch.

EarlyStopping: Training a neural network for too many epochs can cause it to overfit to the training data, hurting its performance on unseen data. The EarlyStopping callback was set to monitor the validation loss, automatically halting training if no improvement was observed over a specified number of epochs (the 'patience' period). As observed in the training logs, this callback stopped the training process at epoch 25 and, crucially, restored the model weights from the best-performing epoch, which was epoch 15. This ensures that the final saved model represents the point of optimal

generalization, not the overfitted state.

ReduceLROnPlateau: During training, the model can reach a plateau where the validation loss stops improving. This can sometimes be resolved by taking smaller steps in the optimization landscape. The ReduceLROnPlateau callback was used to monitor the validation_loss for this purpose. When a plateau was detected, the learning rate was automatically reduced by a predefined factor. For instance, the training logs show that at the end of epoch 20, the learning rate was reduced from 1.0e-03 to 5.0e-04. This dynamic adjustment allows for finer control over the optimization, helping the model to converge to a more optimal minimum

3.6 Evaluation Metrics (MAE, R² Score)

After training, the final performance of both the Random Forest and the deep neural network models was assessed on the unseen test set. This evaluation provides an unbiased measure of how well the models are expected to perform on new, real-world data. Two standard regression metrics were chosen for this purpose.

Mean Absolute Error (MAE):

Mean Absolute Error (MAE) provides a simple and intuitive measure of prediction accuracy, calculated as the average of the absolute differences between predicted and actual values.

$$\text{MAE} = (1 / n) \sum_{i=1}^n |Y_i - \hat{Y}_i|$$

The primary advantage of MAE is its interpretability. Since the metric is in the same units as the target variable (in this case, electron volts, eV), it represents the average absolute error of the model's predictions. For example, an MAE of 10 eV would mean that, on average, the model's energy predictions are off by 10 eV from the true DFT-calculated values. Unlike MSE, MAE does not square the errors, meaning it is less sensitive to outliers and treats all errors on a linear scale. A lower MAE value indicates a more accurate model.

R-squared (R^2) Score: Coefficient of Determination

The R-squared (R^2) score is a statistical measure that represents the proportion of variance in the target variable accounted for by the model's input features, providing a relative indication of the model's goodness of fit. The formula is

$$R^2 = 1 - (SS_{\text{res}} / SS_{\text{tot}}) = 1 - [\sum_{i=1}^n (Y_i - \hat{Y}_i)^2 / \sum_{i=1}^n (Y_i - \bar{Y})^2]$$

Here, SS_{res} is the sum of the squared residuals (the error of our model), and SS_{tot} is the total sum of squares, which represents the error of a simple baseline model that always predicts the mean of the target variable ($\bar{Y}$). The R^2 score therefore compares the performance of our trained model against this naive baseline. The score ranges from $-\infty$ to 1, where:

- $R^2 = 1$: The model perfectly predicts the target variable, explaining 100% of the variance.
- $R^2 = 0$: The model performs no better than simply predicting the mean of the target.
- $R^2 < 0$: The model performs worse than the baseline mean predictor.

For this project, an R^2 score close to 1 is the primary indicator of a successful and highly predictive model.

CHAPTER 4

Results and Discussion

4.1 Experimental Setup

To promote transparency and reproducibility, this section outlines the computational environment, software stack, and data partitioning strategy employed in all experiments.”

Hardware and Software Environment

All data processing, model training, and evaluation were conducted on a standard workstation. The deep neural network training was optimized to leverage available CPU instruction sets such as AVX2 for performance enhancement. The primary programming language used for the project was Python, and the experiments were managed within an interactive development environment.

The software stack was composed of several industry-standard open-source libraries:

- Pandas & NumPy: Used for core data manipulation, including loading the dataset from the CSV file, cleaning the data, and performing numerical operations.
- Scikit-learn: This library was instrumental for multiple stages of the workflow, including data preprocessing (StandardScaler, train_test_split), implementation of the baseline RandomForestRegressor, and calculation of the final evaluation metrics.
- TensorFlow and Keras: The deep neural network was built and trained using the

TensorFlow framework. Keras served as the high-level API for defining the model's architecture, including its layers, activation functions, and callbacks.

Dataset Partitioning

The dataset, consisting of 15,071 samples, was carefully partitioned to ensure a fair and unbiased evaluation of the models' ability to generalize to new data. A standard 80/20 split was employed, resulting in:

- A training set containing 12,056 samples
- A testing set containing 3,015 samples.

The models were trained solely on the training set, while the test set was reserved for the final evaluation of performance after training was completed. A fixed `random_state` was used during the split to ensure that both the Random Forest and the neural network were trained and evaluated on the exact same data partitions, making their performance results directly comparable.

4.2 Performance of the Neural Network Model

The deep neural network (NN), designed with three hidden blocks, was trained to simultaneously predict `Formation_E` and `Total_E`. The training process was managed by `EarlyStopping` and `ReduceLROnPlateau` callbacks, with the detailed epoch-by-epoch training logs available in Appendix A

4.2.1 Quantitative Results

The trained NN model demonstrated exceptionally high predictive accuracy on the unseen test set. The performance, as measured by Mean Absolute Error (MAE) and R-squared (R^2), is summarized below.

Target Property	Metric	Neural Network Performance
Formation Energy	MAE	16.305
	R ² Score	0.997881
Total Energy	MAE	91.187
	R ² Score	0.999445

Table 1: Neural Network Performance Metrics

the R² scores, both exceeding 0.997, indicate that the neural network was able to explain over 99.7% of the variance in the target properties, signifying an excellent model fit.

4.2.2 Visual Analysis of Prediction

The model's high quantitative accuracy is further corroborated by visual analysis. A scatter plot of predicted versus actual Formation_E values for the test set shows data points tightly clustered along the $y = x$ diagonal. This near-perfect linear alignment visually confirms the model's ability to make accurate predictions with minimal error across the full range of energy values.

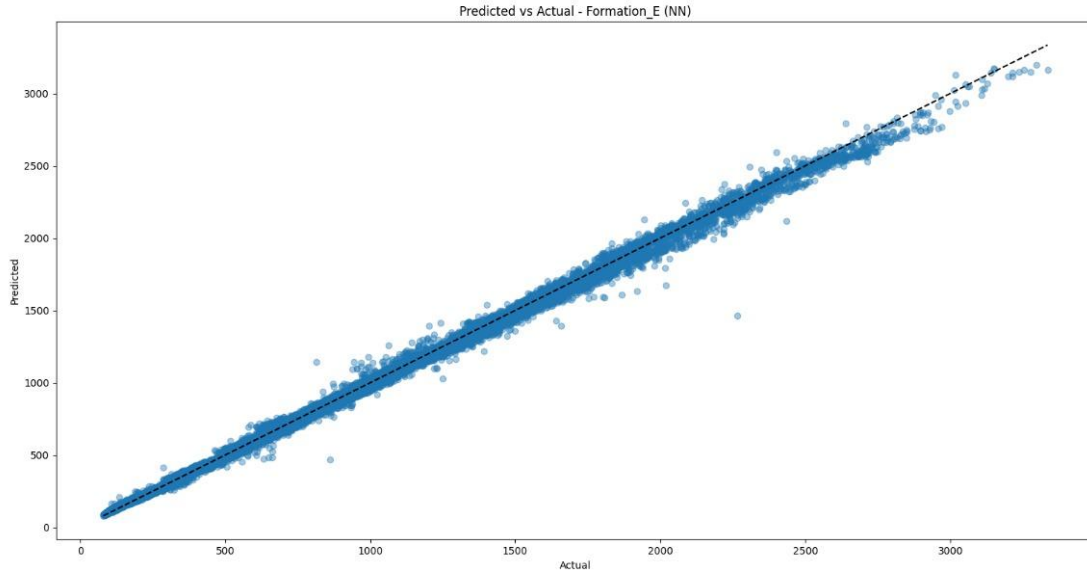


Figure 4.1: Predicted vs. Actual Formation Energy (NN) on the Test Set.

4.3 Performance of the Random Forest Baseline

Real-To establish a high-performance benchmark for this prediction task, (RF) model was implemented and trained on the same dataset. The Random Forest algorithm is a powerful ensemble method renowned for its high accuracy on tabular data, making it an excellent choice for a robust baseline against which the deep neural network can be compared. The model was trained to predict both Formation_E and Total_E, and its performance was evaluated on the unseen test set.

The quantitative performance of the trained Random Forest model is detailed in the table below:

Target Property	Metric	Random Forest Performance
Formation Energy	MAE	4.355
	R² Score	0.999734
Total Energy	MAE	4.708
	R² Score	0.999993

, **Table 2: Random Forest Performance Metrics**

Random Forest model achieved outstanding results, demonstrating an exceptionally strong ability to predict the energetic properties of the nanoparticles. For the Formation Energy, the model yielded a MAE of just 4.355 eV, indicating a very low average prediction error. This was complemented by an R² score of 0.999734, which signifies that the model was able to explain 99.97% of the variance in the formation energy, indicating a near-perfect fit.

The performance for the Total Energy was even more remarkable. The R² score of 0.999993 is extremely close to a perfect score of 1.0. This near-perfect fit indicates that the relationship between the provided structural features and the nanoparticle's total energy is very well-defined and was almost completely captured by the ensemble of decision trees. The exceptional performance of the RF model not only validates the quality and predictive power of the feature set but also establishes a very high benchmark for the comparative analysis in the following section.

4.4 Comparative Analysis of Models

To check the most effective modeling approach for this prediction task, a direct comparative analysis was conducted between the deep neural network (NN) and the Random Forest (RF) baseline. Both models were trained and evaluated on the exact same data partitions, allowing for a fair and direct comparison of their performance on the unseen test set.

The results clearly indicate that while both models achieved exceptionally high levels of performance, the Random Forest model demonstrated superior predictive accuracy across all evaluated metrics. For the prediction of Formation Energy, the Random Forest's MAE of 4.355 eV is significantly lower than the neural network's MAE of 16.305 eV. Similarly, its R^2 score of 0.999734 is markedly closer to a perfect score of 1.0 than the neural network's score of 0.997881.

This trend is even more pronounced for the Total Energy prediction, where the Random Forest's MAE of 4.708 eV is an order of magnitude better than the neural network's 91.187 eV. The R^2 score of 0.999993 for the Random Forest is indicative of an almost flawless fit to the data. While the deep neural network is a highly capable model, these results suggest that for this specific high-dimensional, tabular dataset, the ensemble of decision trees within the Random Forest was more effective at capturing the intricate patterns and relationships in the feature set.

CHAPTER-5

CONCLUSION

This chapter summarizes the key findings of the project and draws final conclusions based on the results of the comparative analysis. It also proposes several promising directions for future research that could build upon the successful framework established in this work

5.1 Conclusion

This his project was initiated to address the significant challenge of the slow and computationally expensive design cycle that currently hinders nanomaterials discovery. The primary objective was to develop and validate a robust, data-driven framework capable of rapidly and accurately predicting the energetic properties of complex bimetallic Cobalt-Gold (CoAu) nanoparticles, thereby serving as an efficient and practical alternative to time-consuming Density Functional Theory (DFT) simulations. To achieve this, a comprehensive methodology was employed, involving the implementation of two powerful but distinct machine learning models—a deep neural network and a Random Forest regressor. These models were trained and evaluated on a high-dimensional dataset of 15,071 unique nanoparticle configurations, each described by over one thousand structural and energetic features.

The results of this study unequivocally demonstrate the success and validity of the machine learning approach for this complex materials science problem. Both implemented models were able to predict the formation energy and total energy of the nanoparticles with exceptionally high accuracy. The deep neural network, representing a state-of-the-art deep learning approach, proved to be a powerful predictor, achieving excellent R-squared (R^2) scores of 0.997881 for Formation Energy and 0.999445 for Total Energy on the unseen test set. The Random Forest model, implemented as a strong baseline, exhibited even higher performance,

achieving a remarkable R^2 score of 0.999734 for Formation Energy and an almost perfect score of 0.999993 for Total Energy.

A key finding from the comparative analysis is the superior performance of the Random Forest model on this specific dataset. This suggests that for highly-structured, tabular data with a well-defined feature set, the ensemble of decision trees in a Random Forest is exceptionally effective at capturing the intricate relationships between a nanoparticle's structure and its energy. While the deep neural network was also highly successful, the Random Forest's ability to achieve a lower prediction error establishes it as the more reliable and accurate model for this particular application.

In conclusion, the primary contribution of this work is the successful validation that a data-driven workflow can replicate the results of computationally expensive simulations with a high degree of accuracy and in a fraction of the time. The project successfully demonstrates that machine learning, particularly with well-suited ensemble models like Random Forest, offers a viable and highly effective methodology for predicting the stability of complex nanoparticles from their structural features alone. This capability has profound implications for materials science, as it enables the high-throughput virtual screening of novel material candidates. By drastically reducing the time required to assess the stability of a new structure, this approach can significantly accelerate the design and discovery pipeline for next-generation materials in fields like catalysis and clean energy.

CHAPTER -6

FUTURE SCOPE

The successful validation of the machine learning models in this project provides a strong foundation for several exciting avenues for future work. The following directions are proposed to enhance the current models' performance, expand their practical usability, and broaden their scientific impact.

6.1 Hyperparameter Tuning and Optimization

Although the models in this study achieved high accuracy, they were trained using baseline hyperparameters. The performance of both Random Forest and, particularly, Deep Neural Networks is highly sensitive to these settings. A thorough hyperparameter tuning could further enhance their predictive capabilities.

For the Random Forest model, this involves systematically adjusting parameters such as `n_estimators` (number of trees), `max_depth` (maximum depth of each tree), and `max_features` (number of features considered at each split). For the deep neural network, a more extensive search could cover the network architecture (number of hidden layers and neurons), dropout rate, activation functions, and the learning rate of the Adam optimizer.

Advanced tuning methods—such as Grid Search, Random Search, or more efficient approaches like Bayesian Optimization—can be employed to systematically explore the hyperparameter space. The ultimate objective of this optimization is to further reduce prediction error and investigate whether a finely tuned neural network can match or surpass the performance of the Random Forest model..

6.2 Deployment as a Predictive Tool

To maximize the impact of this research and make its findings accessible to the broader

materials science community, a key future direction is the deployment of the trained model as a practical, interactive tool. This would involve creating a user-friendly web application or a simple graphical user interface (GUI) using modern Python frameworks such as Streamlit or Flask.

The deployed tool would allow a user, such as an experimental chemist or materials scientist, to input the structural feature vector of a new, hypothetical CoAu nanoparticle. The application would then use the pre-trained, high-performance Random Forest model to provide an instantaneous prediction of its Formation Energy and Total Energy. This would transform the project from a static research study into a dynamic tool for on-the-fly hypothesis testing, enabling researchers who are not machine learning experts to benefit from its predictive power.

6.3 Finalizing and Extending Advanced Models

A significant direction for future work is the completion and formal evaluation of the Graph Neural Network (GNN) model that is currently under development. This would involve:

Finalizing the GNN architecture and conducting a thorough hyperparameter tuning process.

Performing a direct, quantitative comparison of the GNN's performance against the Random Forest and the feature-based neural network. This would provide valuable insights into the trade-offs between using hand-crafted features versus end-to-end representation learning for this problem.

Furthermore, the overall framework could be extended to predict a wider range of material properties. If datasets with additional labels were available, the models could be adapted—for instance, by using a classification head on the neural network—to predict categorical properties such as Thermodynamic Stability (e.g., stable vs. unstable) or Surface Activity for catalysis (e.g., active vs. inactive), making the predictive tool even more comprehensive.

task.

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